S01 :RYU CONTROLLER STARTUP



Figure 1: Successful initialization of the Ryu SDN controller and Smart CDN application, including virtual service initialization and switch connection.

S02 : MININET INITIALISATION

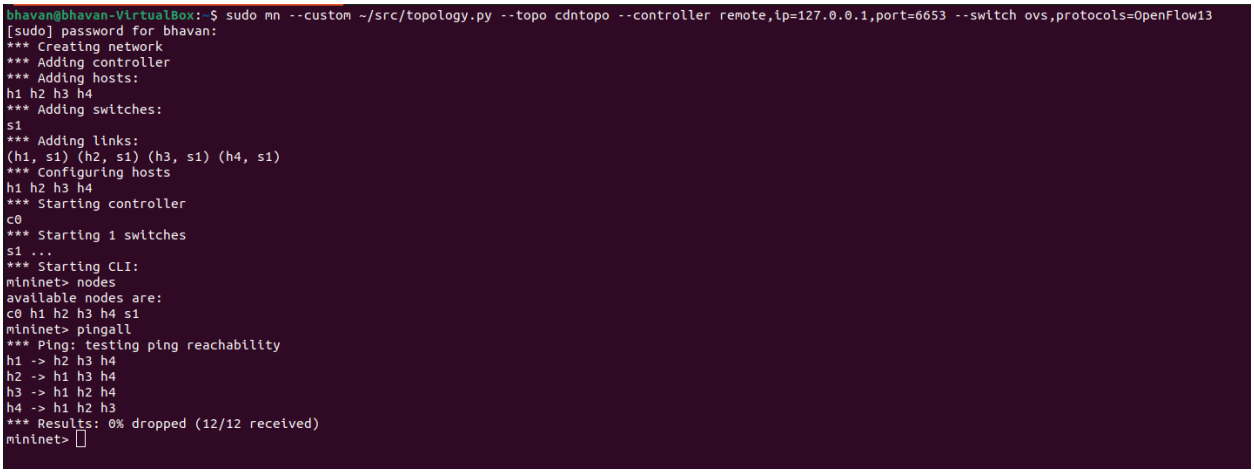


Figure 2: Mininet topology initialization and successful connectivity verification among all four hosts, with zero packet loss in the pingall test.

S03 - START CDN A AND CDN B

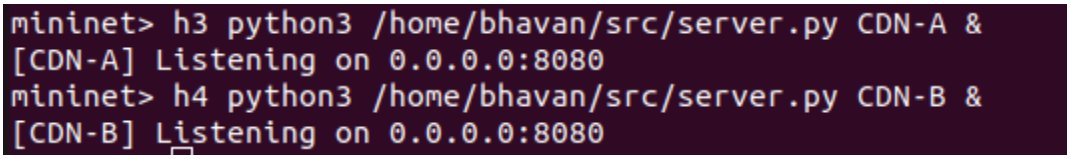


Figure 3: Successful startup of CDN-A and CDN-B content servers on their respective Mininet hosts.

```
bhavan@bhavan-VirtualBox:~$ sudo python3 ~/src/monitor.py
[sudo] password for bhavan:
[10:53:24] [MONITOR] probing from h1; interval=1.0s fail>=2 recover>=2
[10:53:25] [HEALTH] h3 (10.0.0.3) = HEALTHY (detected 1.4s after first good probe)
[10:53:26] [HEALTH] h4 (10.0.0.4) = HEALTHY (detected 1.3s after first good probe)
```

Figure 4: Health monitor checking CDN-A and CDN-B at one-second intervals and reporting both backend servers as healthy.

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request; print(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status', timeout=3).read().decode()
)"
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "roundrobin",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.468065855223237,
      "last_health_check": "2026-10-23 10:55:49",
      "selections": 0
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.410397048398632,
      "last_health_check": "2026-10-23 10:55:50",
      "selections": 0
    }
  }
}
bhavan@bhavan-VirtualBox:~$
```

Figure 5: Smart CDN controller status showing the virtual service at 10.0.0.100:8080, the Round-Robin selection policy, and both CDN backends enabled, healthy, and eligible for selection.

Round robin trial1

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 262.59 ms
Average response throughput  : 20580.87 kbps
Average body throughput      : 20528.36 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

RR TRIAL 2

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 127.23 ms
Average response throughput  : 12626.93 kbps
Average body throughput      : 12594.72 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

RR TRIAL 3

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 354.27 ms
Average response throughput  : 10691.55 kbps
Average body throughput      : 10664.27 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

Figure 6: Succee 6: Successful end-to-end content delivery through the Smart CDN virtual IP using Round-Robin selection, with 10 out of 10 successful HTTP responses.

Metric	Trial 1	Trial 2	Trial 3	Average
Successful responses	10/10	10/10	10/10	30/30 overall
Average response time	262.59 ms	127.23 ms	354.27 ms	**248.03 ms**
Response throughput	20,580.87 kbps	12,626.93 kbps	10,691.55 kbps	**14,633.12 kbps**
Body throughput	20,528.36 kbps	12,594.72 kbps	10,664.27 kbps	**14,595.78 kbps**
Backend distribution	5 A, 5 B	5 A, 5 B	5 A, 5 B	Even distribution in every trial

I**nterpretation:** Round-Robin distributed the ten requests evenly between CDN-A and CDN-B in all three trials. Response time varied considerably, which is why we'll compare averages across repeated runs rather than relying on a single result.

STATIC:

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request; req=urllib.request.Request('http://127.0.0.1:8080/smartcdn/policy/static', data=b'{}', headers={
'Content-Type':'application/json'}, method='POST'); print(urllib.request.urlopen(req, timeout=3).read().decode())"
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "static",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 4.918516200335024,
      "last_health_check": "2026-10-23 11:07:52",
      "selections": 15
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.046797889475098,
      "last_health_check": "2026-10-23 11:07:51",
      "selections": 15
    }
  }
}
```

STATIC TRIAL 1:

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 61.22 ms
Average response throughput : 22623.95 kbps
Average body throughput      : 22566.23 kbps
Servers observed             : CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A
```

TRIAL 2 :

```
==========

SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 151.50 ms
Average response throughput : 17417.42 kbps
Average body throughput      : 17372.98 kbps
Servers observed             : CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A
```

TRIAL 3:

```
==========

SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 212.41 ms
Average response throughput : 11869.51 kbps
Average body throughput      : 11839.23 kbps
Servers observed             : CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A, CDN-A
```

Metric	Trial 1	Trial 2	Trial 3	Average
Successful responses	10/10	10/10	10/10	**30/30**
Average response time	61.22 ms	151.50 ms	212.41 ms	**141.71 ms**
Response throughput	22,623.95 kbps	17,417.42 kbps	11,869.51 kbps	**17,303.63 kbps**
Body throughput	22,566.23 kbps	17,372.98 kbps	11,839.23 kbps	**17,259.48 kbps**

Backend distribution	10 CDN-A	10 CDN-A	10 CDN-A	**30 CDN-A**

Observation: Static selection consistently sent all requests to CDN-A. Its average response time was lower than Round-Robin in these particular runs, but we'll interpret the comparison only after collecting Adaptive results as well.

ADAPTIVE POLICY

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request; req=urllib.request.Request('http://127.0.0.1:8080/smartcdn/policy/adaptive', data=b'{}', headers
={'Content-Type':'application/json'}, method='POST'); print(urllib.request.urlopen(req, timeout=3).read().decode())"
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "adaptive",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.19011096513697,
      "last_health_check": "2026-10-23 11:14:50",
      "selections": 45
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 6.329092682327091,
      "last_health_check": "2026-10-23 11:14:50",
      "selections": 15
    }
  }
}
```

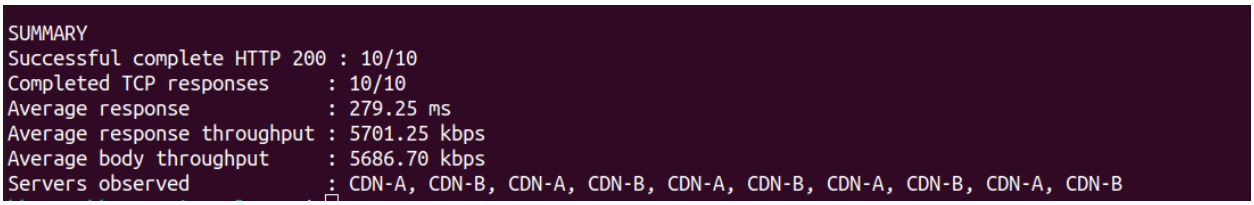
ADAPTIVE TRIAL 1:

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 60.34 ms
Average response throughput  : 25754.36 kbps
Average body throughput      : 25688.66 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

ADAPTIVE TRIAL 2:

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 235.43 ms
Average response throughput  : 11885.16 kbps
Average body throughput      : 11854.84 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

ADAPTIVE TRIAL 3:



Policy	Trial	Successes	Avg. response (ms)	Response throughput (kbps)	Body throughput (kbps)
Round-Robin	1	10/10	262.59	20,580.87	20,528.36
Round-Robin	2	10/10	127.23	12,626.93	12,594.72
Round-Robin	3	10/10	354.27	10,691.55	10,664.27
Static	1	10/10	61.22	22,623.95	22,566.23
Static	2	10/10	151.50	17,417.42	17,372.98
Static	3	10/10	212.41	11,869.51	11,839.23
Adaptive	1	10/10	60.34	25,754.36	25,688.66
Adaptive	2	10/10	235.43	11,885.16	11,854.84
Adaptive	3	10/10	279.25	5,701.25	5,686.70

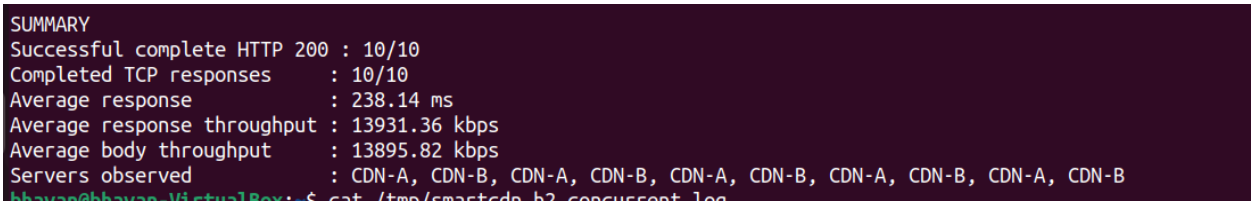
Metric	Round-Robin	Static	Adaptive
Total successful requests	30/30	30/30	30/30
Success rate	100%	100%	100%
Average response time	248.03 ms	**141.71 ms**	191.67 ms
Average response throughput	14,633.12 kbps	**17,303.63 kbps**	14,446.92 kbps
Average body throughput	14,595.78 kbps	**17,259.48 kbps**	14,410.07 kbps

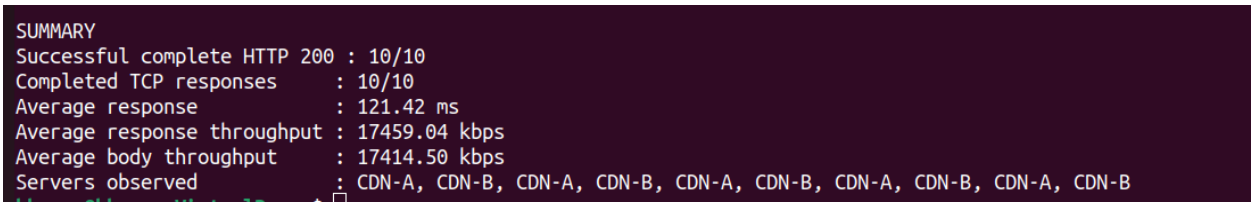
Static performed best on average in these particular trials. It recorded the lowest average response time and highest average throughput among the three policies.

Round-Robin distributed the requests evenly between CDN-A and CDN-B in every trial. Static sent all requests to CDN-A in each recorded trial. Adaptive also alternated between the two backends in these runs.

However, response time and throughput varied considerably between trials. These are measurements from a small, controlled Mininet experiment, not a production CDN benchmark. We'll discuss that variability and the limitations honestly in the report rather than claiming that Adaptive is the fastest.

Concurrent client:

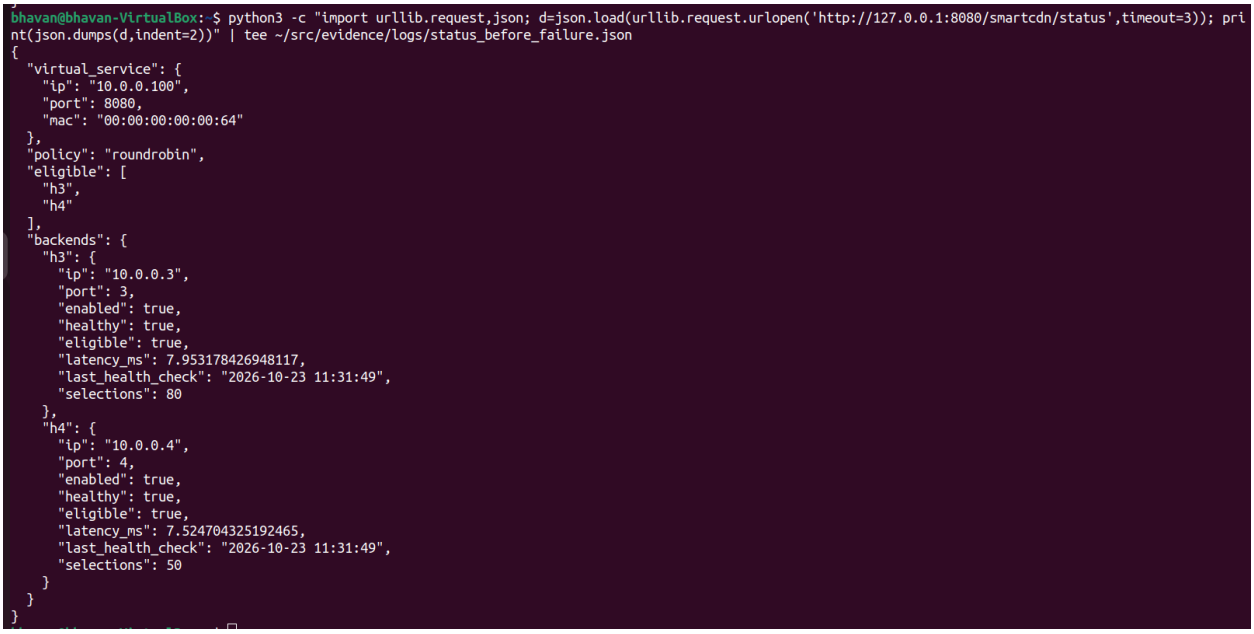




Metric	Client h1	Client h2
Successful HTTP responses	10/10	10/10
Completed TCP responses	10/10	10/10
Average response time	238.14 ms	121.42 ms
Average response throughput	13,931.36 kbps	17,459.04 kbps
Average body throughput	13,895.82 kbps	17,414.50 kbps
Result	**PASS**	**PASS**

Figure 15: Concurrent-client experiment demonstrating successful content delivery to hosts h1 and h2 through the Smart CDN virtual service, with 10 successful HTTP responses from each client.

FINAL RECOVERY TEST :



AUTOMATIC FAILURE DETECTION CYCLE :

CYCLE 1:

ON RUNNING :

h3 pkill -TERM -f '^python3 /home/bhavan/src/server.py CDN-A$'

AND

python3 -c "import urllib.request,json;
d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3));
print(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_h3_down_cycle1.json

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3)); pri
nt(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_h3_down_cycle1.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "roundrobin",
  "eligible": [
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": false,
      "eligible": false,
      "latency_ms": null,
      "last_health_check": "2026-10-23 11:34:20",
      "selections": 80
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.902384092408038,
      "last_health_check": "2026-10-23 11:36:05",
      "selections": 50
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 5/5
Completed TCP responses      : 5/5
Average response             : 263.96 ms
Average response throughput  : 13964.12 kbps
Average body throughput      : 13928.50 kbps
Servers observed             : CDN-B, CDN-B, CDN-B, CDN-B, CDN-B
```

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3)); pri
nt(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_recovered_cycle1.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "roundrobin",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 7.700987228393555,
      "last_health_check": "2026-10-23 11:38:55",
      "selections": 80
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.475696540327174,
      "last_health_check": "2026-10-23 11:38:55",
      "selections": 55
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 181.58 ms
Average response throughput  : 17371.97 kbps
Average body throughput      : 17327.65 kbps
Servers observed             : CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A
```

Cycle 2

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3)); pri
nt(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_h3_down_cycle2.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "roundrobin",
  "eligible": [
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": false,
      "eligible": false,
      "latency_ms": null,
      "last_health_check": "2026-10-23 11:43:00",
      "selections": 85
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.339534743014688,
      "last_health_check": "2026-10-23 11:43:06",
      "selections": 60
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 5/5
Completed TCP responses      : 5/5
Average response             : 885.55 ms
Average response throughput  : 6943.00 kbps
Average body throughput      : 6925.29 kbps
Servers observed             : CDN-B, CDN-B, CDN-B, CDN-B, CDN-B
```

```
bhavan@bhavan-VirtualBox:~$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3)); pri
nt(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_recovered_cycle2.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "roundrobin",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.7143125,
      "last_health_check": "2026-10-23 11:47:15",
      "selections": 85
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.223018658881607,
      "last_health_check": "2026-10-23 11:47:15",
      "selections": 65
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 280.96 ms
Average response throughput  : 3289.65 kbps
Average body throughput      : 3281.26 kbps
Servers observed             : CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B
```

ACTIVATE ADAPTIVE POLICY:

```
bhavan@bhavan-VirtualBox:~/src$ python3 -c "import urllib.request; req=urllib.request.Request('http://127.0.0.1:8080/smartcdn/policy/adaptive', data=b'{}', hea
ders={'Content-Type':'application/json'}, method='POST'); print(urllib.request.urlopen(req, timeout=3).read().decode())"
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "adaptive",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.438522019101997,
      "last_health_check": "2026-10-23 12:07:36",
      "selections": 90
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 10.034599731675467,
      "last_health_check": "2026-10-23 12:07:36",
      "selections": 70
    }
  }
}
```

```
bhavan@bhavan-VirtualBox:~/src$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3));
 print(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_adaptive_h3_down.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "adaptive",
  "eligible": [
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": false,
      "eligible": false,
      "latency_ms": null,
      "last_health_check": "2026-10-23 12:09:54",
      "selections": 90
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 7.809119333579423,
      "last_health_check": "2026-10-23 12:10:01",
      "selections": 70
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 5/5
Completed TCP responses      : 5/5
Average response             : 267.61 ms
Average response throughput  : 23141.72 kbps
Average body throughput      : 23082.68 kbps
Servers observed             : CDN-B, CDN-B, CDN-B, CDN-B, CDN-B
```

```
bhavan@bhavan-VirtualBox:~/src$ python3 -c "import urllib.request,json; d=json.load(urllib.request.urlopen('http://127.0.0.1:8080/smartcdn/status',timeout=3));
 print(json.dumps(d,indent=2))" | tee ~/src/evidence/logs/status_adaptive_recovered.json
{
  "virtual_service": {
    "ip": "10.0.0.100",
    "port": 8080,
    "mac": "00:00:00:00:00:64"
  },
  "policy": "adaptive",
  "eligible": [
    "h3",
    "h4"
  ],
  "backends": {
    "h3": {
      "ip": "10.0.0.3",
      "port": 3,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 8.943125,
      "last_health_check": "2026-10-23 12:13:05",
      "selections": 90
    },
    "h4": {
      "ip": "10.0.0.4",
      "port": 4,
      "enabled": true,
      "healthy": true,
      "eligible": true,
      "latency_ms": 9.466494260651917,
      "last_health_check": "2026-10-23 12:13:05",
      "selections": 75
    }
  }
}
```

```
SUMMARY
Successful complete HTTP 200 : 10/10
Completed TCP responses      : 10/10
Average response             : 233.24 ms
Average response throughput  : 22655.89 kbps
Average body throughput      : 22598.09 kbps
Servers observed             : CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A, CDN-B, CDN-A
```

Metric	Result
Successful HTTP responses	10/10
Completed TCP responses	10/10
Average response time	233.24 ms
Average response throughput	22,655.89 kbps
Average body throughput	22,598.09 kbps
Backends observed	CDN-B and CDN-A
Result	**PASS**

FINAL RESULTS:

Policy	Average response time	Average throughput	Successful requests
Round-robin	248.03 ms	14,633.12 kbps	30/30
Static	141.71 ms	17,303.63 kbps	30/30
Adaptive	191.67 ms	14,446.92 kbps	30/30